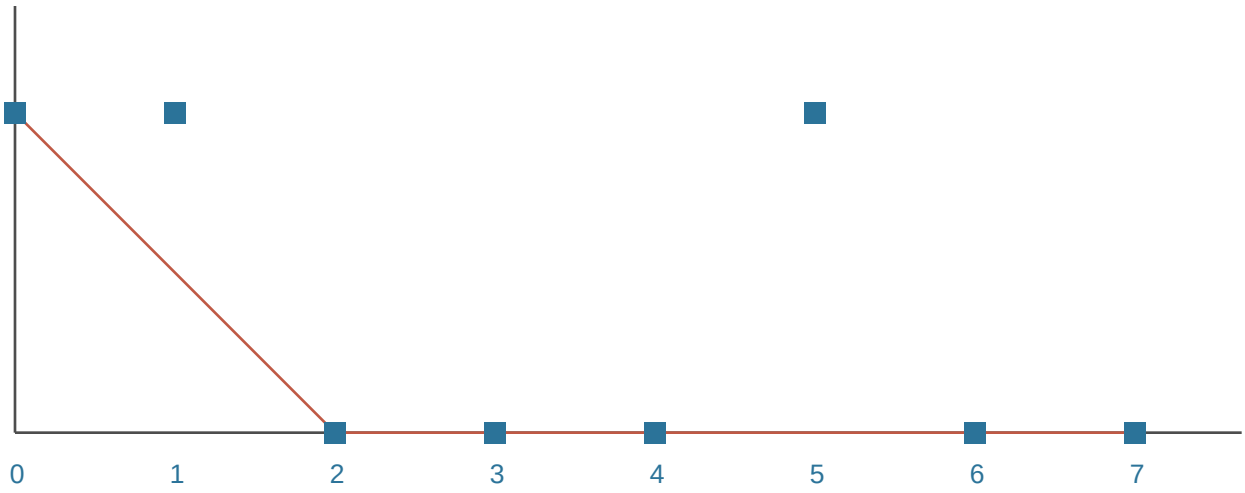
R128: Newton expansion at the degree-7 double root

F(s,u)=f23(a+s,90+u) over F128; 52 nonzero terms

First s-orders by u-degree: 1,1,0,0,0,1,0,0

Lower hull: (0,1) -> (2,0) -> (7,0)

Positive edge residual: 84 + 110 Z^2; slope -1/2



Transverse derivative coefficient: 84 != 0 in F128; 84*6 = 110.

Next: lift s(u) directly, then certify normalized ramification and lambda branches.

Checks: independent Horner, byte replay, changed-source rejection, Lean finite claims.

Result SHA-256: cce71416b01d600abbeedbd86f56f647c6c91b7d2ee536d6f730d5e9c79a739d